University of Illinois
ECE 310 **Profs. Do, Shomorony, Snyder**

Midterm Exam 1

7:00-9:00pm, Wednesday, October 2, 2024

Name: _____

Section: 9:00 AM 12:00 PM 3:00 PM

NetID: _____

Score: _____

Problem	Pts.	Score
1	12	
2	12	
3	8	
4	10	
5	15	
6	15	
7	18	
8	10	
Total	100	

Instructions

- You may not use any books, calculators, or notes other than one handwritten two-sided sheet of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

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(12 Pts.)

1. Select **True** or **False** to each of the following statements:

   (a) For a discrete-time system with impulse response $h[n]$, the system output $y[n]$ to any input signal $x[n]$ is always determined using $h[n]$. **True/<u>False</u>**

   (b) The impulse response $h[n]$ of an unstable LTI system will always be an unbounded sequence. **True/<u>False</u>**

   (c) Two BIBO stable LTI systems connected in parallel will always form a BIBO stable LTI system. **<u>True</u>/False**

   (d) An LCCDE system can only have a finite number of poles. **<u>True</u>/False**

   (e) An LTI system given by $y[n] = y[n-3] + x[n]$ will have three distinct poles in its transfer function. **<u>True</u>/False**

   (f) An LTI system with transfer function $H(z) = \frac{1}{1-0.5z^{-1}} + \frac{1}{1-2z^{-1}}$ must **not** be BIBO stable. **True/<u>False</u>**

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(12 Pts.)

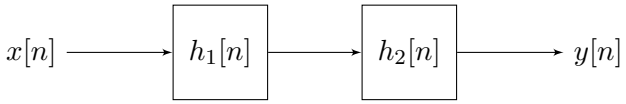
2. For each of the systems with input $x[n]$ and output $y[n]$ shown in the table, indicate by "**yes**" or "**no**" whether the properties indicated apply to the system.

	Linear	Shift-Invariant	Causal	Stable
$y[n] = x[n]\ x[n+1]$	**No**	**Yes**	**No**	**Yes**
$y[n] = \frac{1}{\lvert n\rvert+1} x[n]$	**Yes**	**No**	**Yes**	**Yes**
$y[n] = \sin(x[n]) + x[0]$	**No**	**No**	**No**	**Yes**

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(8 Pts.)

3. Consider a system composed of two LTI systems connected in series.



The input $x[n] = \{\underset{\uparrow}{1}, -1\}$ produces output $y[n] = \{4, \underset{\uparrow}{-2}, -2\}$. We also know that $h_2[n] = \{\underset{\uparrow}{2}, 1\}$.

(a) Determine $h_1[n]$.

First, we have that $y[n] = x[n] * h_1[n] * h_2[n]$. By commutativity of convolution, we have $y[n] = x[n] * h_2[n] * h_1[n]$. Using the available information, we may find

$$\{4, \underset{\uparrow}{-2}, -2\} = \{\underset{\uparrow}{2}, -1, -1\} * h_1[n].$$

Clearly, $y[n]$ is the same as $x[n] * h_1[n]$ shifted left by one sample and multiplied by two. Thus, but the linearity and time-invariance of convolution, we must have:

$$\boxed{h_1[n] = 2\delta[n+1]}$$

(b) Determine the overall system's impulse response $h[n]$ such that $y[n] = x[n] * h[n]$. Is this overall system causal? Justify your reasoning.

The overall system will have impulse response $h[n] = h_1[n] * h_2[n]$.

$$\boxed{h[n] = \{4, \underset{\uparrow}{2}\}}$$

This overall system is **non-causal** because $h[n]$ is not zero for all $n < 0$.

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(10 Pts.)

4. For each of the following parts, compute the convolution $x[n] * h[n]$ for the given sequences.

   (a) $x[n] = \{1, -3, \underset{\uparrow}{2}, -1, 4\},\ h[n] = \{2, \underset{\uparrow}{0}, -1\}$

$$y[n] = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & -3 & 1 \\ -1 & 2 & -3 \\ 4 & -1 & 2 \\ 0 & 4 & -1 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 3 \\ 1 \\ 6 \\ 1 \\ -4 \end{bmatrix}$$

$$\boxed{y[n] = \{2, -6, 3, \underset{\uparrow}{1}, 6, 1, -4\}}$$

   (b) $x[n] = e^{-n}u[n],\ h[n] = (n+1)(u[n] - u[n-3])$

$$h[n] = \{\underset{\uparrow}{1}, 2, 3\}$$

$$x[n] * h[n] = x[n] + 2x[n-1] + 3x[n-2]$$

$$\boxed{x[n] = e^{-n}u[n] + 2e^{-(n-1)}u[n-1] + 3e^{-(n-2)}u[n-2]}$$

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(15 Pts.)

5. For each of the following discrete-time signals, find its z-transform and the corresponding region of convergence.

(a) $x[n] = (n+1)u[n-1]$.

- Method 1

$$x[n] = (n-1)u[n-1] + 2u[n-1]$$

Applying differentiation and delay properties:

$$\boxed{X(z) = \frac{z^{-2}}{(1-z^{-1})^2} + \frac{2z^{-1}}{1-z^{-1}},\ |z| > 1}$$

- Method 2

$$x[n] = nu[n-1] + u[n-1]$$

Applying delay property to obtain the Z-transform of $u[n-1]$ and applying differentiation on $u[n-1]$ we can obtain another way to express the Z transform, which are essentially same as the result obtained by method 1:

$$\boxed{X(z) = \frac{z^{-1}}{(1-z^{-1})^2} + \frac{z^{-1}}{1-z^{-1}},\ |z| > 1}$$

(b) $x[n] = u[n-1]u[3-n]$.

$$x[n] = \delta[n-1] + \delta[n-2] + \delta[n-3]$$

$$\boxed{X(z) = z^{-1} + z^{-2} + z^{-3},\ z \neq 0}$$

(c) $x[n] = 3^{-n}u[n] + 3^n u[-n]$.

$$x[n] = \left(\frac{1}{3}\right)^n u[n] + 3(3)^{n-1}u[-n]$$

$$\boxed{X(z) = \frac{1}{1-\frac{1}{3}z^{-1}} - \frac{3z^{-1}}{1-3z^{-1}},\ \frac{1}{3} < |z| < 3}$$

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(15 Pts.)

6. A causal LTI system produces the output $y[n] = \{1, 1/2, 0, 0, 0, 0, \ldots\}_{n=0}^{\infty}$ when excited by the input signal $x[n] = \{1, 1/3, 0, 0, 0, 0, \ldots\}_{n=0}^{\infty}$.

   (a) Determine the transfer function of the system.

$$Y(z) = 1 + \frac{1}{2}z^{-1}$$
$$X(z) = 1 + \frac{1}{3}z^{-1}$$

$$\boxed{\frac{Y(z)}{X(z)} = H(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 + \frac{1}{3}z^{-1}},\ |z| > \frac{1}{3}}$$

   (b) Determine the impulse response of the system.

$$\boxed{h[n] = \left(-\frac{1}{3}\right)^n u[n] + \frac{1}{2}\left(-\frac{1}{3}\right)^{n-1} u[n-1]}$$

   OR,

$$\boxed{h[n] = \delta[n] + \frac{1}{6}(-\frac{1}{3})^{n-1}u[n-1]}$$

   OR,

$$\boxed{h[n] = \frac{3}{2}\delta[n] - \frac{1}{2}(-\frac{1}{3})^{n}u[n]}$$

   (c) Express the system in the form of an LCCDE.

$$\frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{2}z^{-1}}{1 + \frac{1}{3}z^{-1}}$$
$$Y(z)(1 + \frac{1}{3}z^{-1}) = X(z)(1 + \frac{1}{2}z^{-1})$$

$$\boxed{y[n] = -\frac{1}{3}y[n-1] + x[n] + \frac{1}{2}x[n-1]}$$

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(18 Pts.)

7. A causal LTI system has transfer function given by

$$H(z) = \frac{1}{(1-\frac{1}{2}z^{-1})(1-2z^{-1})}.$$

(a) Sketch the ROC of $H(z)$ and clearly indicate the location of the poles. The two poles are located at $z = \frac{1}{2}$ and $z = 2$. The system is causal, thus, the ROC will be $|z| > 2$. The sketch of the ROC should be the exterior of a circle of radius 2 with the aforementioned poles indicated.

(b) Find the impulse response $h[n]$.

$$\begin{aligned} H(z) &= \frac{1}{(1-\frac{1}{2}z^{-1})(1-2z^{-1})} \\ &= \frac{A}{1-\frac{1}{2}z^{-1}} + \frac{B}{1-2z^{-1}} \\ 1 &= A(1-2z^{-1}) + B(1-\tfrac{1}{2}z^{-1}) \end{aligned}$$

Plugging in each pole, we find $A = -\frac{1}{3}$ and $B = \frac{4}{3}$. Since the system is causal,

$$\boxed{h[n] = -\frac{1}{3}\left(\frac{1}{2}\right)^n u[n] + \frac{4}{3}(2)^n u[n]}$$

(c) Suppose the input to this system is right-sided and has z-transform

$$X(z) = \frac{z^{-1} - 2z^{-2}}{1-\frac{1}{2}z^{-1}}.$$

Find the z-transform $Y(z)$ of the output of the system and its ROC. Check for any pole-zero cancellations that may occur.

$$\begin{aligned} Y(z) &= X(z)H(z) \\ &= \frac{z^{-1}(1-2z^{-1})}{1-\frac{1}{2}z^{-1}} \cdot \frac{1}{(1-\frac{1}{2}z^{-1})(1-2z^{-1})} \end{aligned}$$

$$\boxed{Y(z) = \frac{z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)^2}, \ |z| > \frac{1}{2}}$$

(d) Determine the the output signal $y[n]$ corresponding to $Y(z)$ in part (c). Is $y[n]$ bounded?

Using the z-transform pair for $Y(z) = \frac{az^{-1}}{1-az^{-1}}$, $|z| > |a|$, and the linearity property, we find that $y[n] = 2n\left(\frac{1}{2}\right)^n u[n]$. This signal is bounded because $\left(\frac{1}{2}\right)^n$ decays much faster than n increases. In particular, we notice that $n\left(\frac{1}{2}\right)^n \leq 1$ for $n \geq 0$.

$$\boxed{y[n] = 2n\left(\frac{1}{2}\right)^n u[n],\ y[n] \text{ is bounded.}}$$

(10 Pts.)

8. (a) Suppose we have a **non-causal** LTI system described by:

$$y[n] + \alpha y[n-1] = x[n] + 3x[n-1],$$

where α is a finite, real-valued constant. For what value(s) of α is this system BIBO stable? The transfer function of this system will be

$$\begin{aligned} Y(z)(1+\alpha z^{-1}) &= X(z)(1+3z^{-1} \\ \frac{Y(z)}{X(z)} = H(z) &= \frac{1+3z^{-1}}{1+\alpha z^{-1}},\ |z| < |\alpha|. \end{aligned}$$

Note that the ROC is on the interior of the pole at $-\alpha$ due to the system being non-causal and necessarily left-sided. We need the ROC to contain the unit circle. Therefore, the system will be BIBO stable if

$$\boxed{|a| > 1}$$

(b) Suppose we have a **causal** LTI system described by:

$$y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n] + \beta x[n-1],$$

where β is a finite, real-valued constant. For what value(s) of β is this system BIBO stable? The transfer function of this system will be

$$\begin{aligned} Y(z)\left(1 - \frac{5}{2}z^{-1} + z^{-2}\right) &= X(z)(1+\beta z^{-1}) \\ H(z) = \frac{1+\beta z^{-1}}{1-\frac{5}{2}z^{-1}+z^{-2}} &= \frac{1+\beta z^{-1}}{(1-2z^{-1})\left(1-\frac{1}{2}z^{-1}\right)}. \end{aligned}$$

The ROC of this system will be $|z| > 2$ and thus unstable unless we choose β such that we cancel the pole at $z = 2$. Therefore, the system will be BIBO stable if

$$\boxed{\beta = -2}$$